

Dr Benya Wang

AI & ML Engineer · PhD

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CAREER PROFILE

AI and ML engineer with a PhD and 8 years taking AI from prototype to production across mining and resources. Strongest on the deployment half: model endpoints, containerised microservices, edge inference on disconnected sites, CI/CD, and the MLOps tooling that keeps models running. Delivers Generative AI at enterprise scale through RAG knowledge bases and agentic workflows on Azure (AI Foundry, Azure OpenAI, AI Search, Azure ML) with Databricks and MLflow. Sets the standards, reference architectures, and platform patterns that let delivery teams build the same way, and designs responsible AI governance with Cyber and Enterprise Architecture. Works with complex operational data (time-series, sensor, historian) across Australia, Africa, and South America.

CORE COMPETENCIES

MLOps, Deployment & Responsible AI

Model endpoint deployment, Docker microservices, ACR, MLflow, Azure DevOps / Pipelines, CI/CD design, edge inference (Azure IoT Edge), documented AI workflows, AI governance frameworks

Generative AI & Agentic Systems

RAG pipeline design, agentic and multi-agent workflows, Azure OpenAI, Azure AI Foundry, Azure AI Search, LangChain, LightRAG, GraphRAG, AutoGen, vector stores, Microsoft Copilot Studio, Claude Code

Azure & Data Platforms

Azure (AI Foundry, Azure ML, AI Search, IoT Edge/Hub, ACR), Databricks (Feature Store, workflow orchestration, data warehouse), AWS

Machine Learning & Modelling

Predictive analytics, time-series and sensor data modelling, feature engineering, XGBoost, LightGBM, Random Forest, computer vision (YOLO), advanced process control

Engineering & Delivery

Python, PySpark, Pandas, SQL, stakeholder engagement, technical mentoring, cross-site delivery across multiple time zones

PROFESSIONAL EXPERIENCE

Specialist Machine Learning Engineer

Apr 2023 – Present

South32 · Perth, WA

- Designed and implemented end to end an air-gapped edge ML deployment pattern (Azure IoT Edge, Databricks Apps), enabling model inference on isolated industrial sites with zero internet connectivity; adopted as the standard across all remote operations.
- Built a safety chatbot and GenAI knowledge base over safety and geoscience corpora (Azure AI Foundry, Azure AI Search, LightRAG), covering corpus exploration, chunking strategy, retrieval tuning, and production deployment, giving operational teams conversational access to guidance locked in unstructured documents.
- Applied responsible AI in practice: partnered with Cyber and Enterprise Architecture to design a Microsoft Copilot Studio governance framework and a GitHub / GitHub Copilot governance framework, establishing enterprise-wide standards for AI-assisted development, access controls, and compliant deployment.
- Defined the technical strategy and built a unified AI platform (Azure AI Search, Azure OpenAI, Cosmos DB) with self-service patterns and reference implementations that reduced time-to-production for new LLM applications by ~60%, enabling multiple product teams to ship independently.
- Authored the standards, engineering guides, and platform patterns that made AI delivery repeatable and well documented across the business, and mentored data scientists across business units to drive consistent, production-quality outcomes.

- Owned model deployment and optimisation, right-sizing the container runtime for inference endpoints: ACI for lightweight, single-service workloads and AKS where orchestration and scaling were required, packaging models as Docker microservices published through ACR.
- Served as Acting Data Science Lead May–Jun 2026, maintaining delivery continuity across active AI workstreams; co-designed the production support model with the Operational Digital Technology team, defining ownership, escalation paths, and SLA structures.

Specialist Data Scientist

Jan 2022 – Apr 2023

South32 · Perth, WA

- Architected and led delivery of a centralised Data Science platform (Azure Databricks, MLflow, Azure Pipelines) that standardised MLOps workflows across 6+ business units, cutting the average model deployment cycle from ~3 weeks to under 5 days and eliminating ad-hoc deployment risk.
- Delivered advanced process control solutions over plant time-series and sensor data for operations in Australia, Africa, and South America, working with local technology teams in each region to improve recovery and energy efficiency, coordinating delivery across multiple time zones.
- Built reusable PySpark feature pipelines into a Databricks Feature Store over large-scale industrial sensor and historian data, replacing ad-hoc notebook preparation with versioned, model-ready features.
- Implemented production multi-agent LLM pipelines (GraphRAG, AutoGen) to extract structured insights from HR records, safety incident reports, and geoscience exploration corpora.
- Designed and deployed a Vision AI safety system (YOLO, Supervision) for automated stop-sign monitoring, directly improving operational safety compliance.

Data Scientist

Jan 2021 – Jan 2022

Rio Tinto · Perth, WA

- Engineered serverless data pipelines (AWS Lambda, Glue, Athena) for real-time production analytics supporting operational decision-making.
- Led ML-driven optimisation of ore blending strategies, improving resource utilisation and reducing annual operational costs.

Graduate Data Scientist

Feb 2019 – Jan 2021

Rio Tinto · Perth, WA

- Developed predictive maintenance models (LightGBM, Random Forest) for a 1,700 km rail network, reducing unplanned downtime through early fault detection.
- Containerised models in Docker and deployed via AWS SageMaker, establishing repeatable deployment and rollback processes for the team.

EDUCATION

PhD in Environmental Engineering

Feb 2015 – Feb 2019

University of Western Australia

Thesis: Application of machine learning algorithms in urban catchments. Five peer-reviewed publications in *Geoscientific Model Development*, *Water Resources Research*, and other journals.

Bachelor of Science

Sep 2010 – Jun 2014

Zhejiang University, China

CERTIFICATIONS

Databricks Certified

Machine Learning Professional · Machine Learning Associate · Data Engineer Associate

Microsoft Certified

DevOps Engineer Expert · Azure Data Scientist Associate · Azure Data Engineer Associate · Azure Developer Associate

AWS Certified

Machine Learning – Specialty · Solutions Architect – Associate

Other

HashiCorp Certified: Terraform Associate · Anthropic: Claude Code in Action